

ASSIGNMENT-1

Q1. What is a variable in C/C++?

Ans. A variable is a named memory location used to store data.

Q2. Why do we use variables in programs?

Ans. We use variables to store and manipulate data in programs.

Q3. What is the difference between a variable name and its value?

Ans. A variable name is the identifier, while its value is the data stored in it.

Q4. Which of the following is a valid variable name?

a) 2num b) total_marks c) for d) my marks

Ans. b) total_marks is the valid variable name.

Q5. Why is Age different from age?

Ans. Because C/C++ is case-sensitive, Age and age are treated as different variables.

Q6. Can a variable name contain spaces? Why?

Ans. No, variable names cannot contain spaces because spaces separate different tokens.

Q7. Can a variable name start with an underscore () ?

Ans. Yes, a variable name can start with an underscore ().

Q8. What happens if two variables have the same name in the same scope?

Ans. It causes a compilation error.

Q9. Write three meaningful variable names for storing student information.

Ans. studentName, rollNumber, and studentMarks.

Q10. Which naming convention is better and why: studentMarks or sm?

Ans. studentMarks is better because it is more meaningful and easier to understand

Q11. What is a data type?

Ans. A data type defines the kind of data a variable can store.

Q12. Which data type is used to store whole numbers?

Ans. int is used to store whole numbers.

Q13. Which data type is used to store decimal values?

Ans. float or double is used to store decimal values

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Q14. What is the difference between float and double?

Ans. double provides more precision and storage than float.

Q15. Which data type stores a single character?

Ans. char is used to store a single character.

Q16. Which data type stores True/False values?

Ans. bool is used to store True/False values.

Q17. What data type is suitable for storing a person's name?

Ans. string is suitable for storing a person's name.

Q18. Predict the output: `int age = 18; cout << age;`

Ans. The output will be 18.

Q19. Identify the data type: 25, 3.14, 'A', true.

Ans. 25 → int, 3.14 → double, 'A' → char, true → bool.

Q20. What happens if you store a decimal number in an int variable?

Ans. The decimal part is discarded, and only the whole number is stored.

Q21. What is the purpose of cin?

Ans. cin is used to take input from the user.

Q22. What is the purpose of cout?

Ans. cout is used to display output on the screen.

Q23. Which symbol is used with cin?

Ans. The >> symbol is used with cin.

Q24. Which symbol is used with cout?

Ans. The << symbol is used with cout.

Q25. Explain: `cin >> age;`

Ans. It takes a value entered by the user and stores it in the variable age.

Q26. Explain: `cout << age;`

Ans. It displays the value of the variable age on the screen.

Q27. What is the purpose of endl?

Ans. endl moves the cursor to the next line.

Q28. Write a statement to input a student's marks.

Ans. `cin >> marks;`

Q29. Write a statement to display "Welcome to C++".

Ans. `cout << "Welcome to C++";`

Q30. Predict the output: `cout << "Programming";`

Ans. The output will be Programming.

Q31. What is an operator?

Ans. An operator is a symbol used to perform operations on data.

Q32. Which operator is used for addition?

Ans. The + operator is used for addition.

Q33. What is the result of `10 % 3`?

Ans. The result is 1. This (%) operator gives remainder.

Q34. What is the result of `8 % 2`?

Ans. The result is 0.

Q35. Which operator checks equality?

Ans. The == operator checks equality.

Q36. What is the difference between = and ==?

Ans. = is used for assignment, while == is used for comparison.

Q37. What is the result of $5 > 3$?

Ans. The result is true.

Q38. What does && represent?

Ans. && represents the Logical AND operator.

Q39. What does || represent?

Ans. || represents the Logical OR operator.

Q40. What does ! represent?

Ans. ! represents the Logical NOT operator.

Q41. Write a program to print "Hello World".

1. Problem Statement:

Write a program to print "Hello World".

2. Algorithm/Steps:

Start.

Display "Hello World".

Stop.

3. Dry Run with Sample Input:

No input required.

4. Program code(C++):

```
#include<iostream>

using namespace std;

int main(){

cout << "Hello World";

return 0;

}
```

Output-

Hello World

=== Code Execution Successful ===

6. Explanation of the Logic Used:

The program uses cout to display "Hello World" on the screen.

Q42. Write a program to input two numbers and display their sum.

1. Problem Statement:

Write a program to input two numbers and display their sum.

2. Algorithm/Steps:

Start.

Input two numbers.

Calculate their sum.

Display the sum.

Stop.

3. Dry Run with Sample Input:

Input: 45 ,54

Sum = 99

4.program code:

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int a, b;
```

```
    cout << "enter two number: ";
```

```
    cin >> a >> b;
```

```
    cout << "sum = " << a + b ;
```

```
    return 0;
```

```
}
```

Ouput-

```
enter two number:45
```

```
54
```

```
sum=99
```

```
=== Code Execution Successful ===
```

6. Explanation of the Logic Used:

The program takes two numbers as input, adds them, and displays the result.

Q43. Write a program to calculate the area of a rectangle.

1. Problem Statement:

Write a program to calculate the area of a rectangle.

2. Algorithm/Steps:

- 1.Start.
- 2.Input length and breadth.
- 3.Calculate Area = Length × Breadth.
- 4.Display the area.
- 5.Stop.

3. Dry Run with Sample Input:

Length = 5, Breadth = 6

Area = 30

Program code:

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {  
    int length, breadth;  
    cout << "enter length and breadth ";  
    cin >> length >> breadth;  
    cout << "Area = " << length * breadth << endl;  
  
    return 0;  
}
```

Output-

```
enter length and breadth 5  
6  
Area = 30
```

```
=== Code Execution Successful ===
```

6. Explanation of the Logic Used:

The program multiplies the length and breadth to find the area of the rectangle.

Q44. What formula is used to calculate Simple Interest?

Ans **$SI = (P \times R \times T) / 100$**

Where:

SI = Simple Interest

P = Principal Amount (the original amount of money)

R = Rate of Interest (percentage per year)

T = Time (in years)

Q45. Write a program to calculate Simple Interest.

1. Problem Statement:

Write a program to calculate Simple Interest.

2. Algorithm/Steps:

Start.

Input Principal, Rate, and Time.

Calculate $SI = (P \times R \times T) / 100$.

Display SI.

Stop.

3. Dry Run with Sample Input:

P = 1000, R = 5, T = 2

$SI = (1000 \times 5 \times 2) / 100$

SI = 100

4. Program Code (C++):

```
#include<iostream>

using namespace std;

int main()
{
    float P, R, T, SI;
    cout << "Enter Principal Amount (P): ";
    cin >> P;
    cout << "Enter Rate of Interest (R): ";
    cin >> R;
    cout << "Enter Time in years (T): ";
    cin >> T;
    SI = (P * R * T) / 100;
    cout << "Simple Interest = " << SI << endl;
    return 0;
}
```

Output –

```
Enter Principal Amount (P): 1000
Enter Rate of Interest (R): 5
Enter Time in years (T): 2
Simple Interest = 100
```

=== Code Execution Successful ===

Q46. Why is a temporary variable used while swapping two numbers?

Ans. A temporary variable is used to store one value temporarily so that it is not lost during swapping.

Q47. Swap the values: a = 10, b = 20.

1. Problem Statement:

Write a program to swap the values of two variables using a temporary variable.

2. Algorithm/Steps:

- 1.Start.
- 2.Declare variables a, b, and temp.
- 3.Assign a = 10 and b = 20.
- 4.Store the value of a in temp.
- 5.Assign the value of b to a.
- 6.Assign the value of temp to b.
- 7.Display the swapped values.
- 8.Stop.

3. Dry Run with Sample Input:

Steps	a	b	temp
Initial	10	20	-
Temp=a	10	20	10
a=b	20	20	10
b=temp	20	10	10

program code-

```
#include<iostream>

using namespace std;

int main()
{
    int a = 10, b = 20, temp;

    temp = a;
    a = b;
    b = temp;

    cout << "After swapping:" << endl;
    cout << "a = " << a << endl;
    cout << "b = " << b << endl;

    return 0;
}
```

Output-

After swapping:

a = 20

b = 10

6.

=== Code Execution Successful ===

Explanation of the Logic Used:

The program uses a temporary variable temp to store the value of a so that it is not lost while exchanging the values of a and b.

Q48. What formula is used to convert Celsius into Fahrenheit?

Ans. Fahrenheit = (Celsius × 9/5) + 32

Q49. What is the Fahrenheit value of 0°C?

Ans. The Fahrenheit value of 0°C is 32°F.

Q50. Why can integer division give incorrect results in the Celsius-to-Fahrenheit program?

Ans. Integer division discards the decimal part, which can lead to inaccurate Fahrenheit values.

